

SECTION 1: Simulation and Modeling (CT 753)

Paper 1: 2082 Shrawan

Attempt all questions.

1. [Q1] Define system, model, and simulation. Differentiate between continuous and discrete event simulation with suitable examples.
2. [Q2] Describe the components of a discrete-event system simulation. Explain the event scheduling algorithm using a flow chart.
3. [Q3] What is random number generation? Explain the linear congruential method for generating pseudo-random numbers with an example.
4. [Q4] Differentiate between parametric and non-parametric tests. Explain the Chi-square goodness-of-fit test procedure in detail.
5. [Q5] A user owning a Pulsar today has an 80% chance of buying a Pulsar again and a 20% chance of switching to Yamaha. A user owning a Yamaha today has an 85% chance of buying a Yamaha again and a 15% chance of switching to Pulsar.
 - o (a) If a user owns a Yamaha today, what is the probability that they will buy a Pulsar after 3 purchases?
 - o (b) If 60% of users own a Pulsar today, what percentage of users will own a Pulsar after 3 purchases?
6. [Q6] A sequence of 10,000 5-digit numbers generated shows the following observed frequencies (\$O_i\$):
 - o All different digits: 3054
 - o One pair: 5020
 - o Two pairs: 1073
 - o Three of a kind: 710
 - o Full house: 95
 - o Four of a kind: 44
 - o Five of a kind: 4

Test independence at $\alpha = 0.05$ (Tabular value $\chi^2 = 12.59$).

Paper 2: 2082 Chaitra

Attempt all questions.

1. [Q1] What are the steps involved in a simulation study? Draw a neat block diagram showing the overall simulation process.
2. [Q2] Explain inverse transform technique for generating continuous random variates. Derive the generator formula for an exponential distribution.
3. [Q3] Differentiate between verification and validation of simulation models. Describe the techniques used to validate a simulation model.
4. [Q4] Describe queueing notation $M/M/1$ and $M/G/1$. Explain Little's Law and its significance in queueing systems.
5. [Q5] Write short notes on:

- (a) High-level architecture (HLA) in simulation
 - (b) Input modeling and distribution fitting
6. [Q6] A sequence of 10,000 5-digit numbers was recorded with the following results:
- All different: 3054
 - One pair: 5020
 - Two pairs: 1073
 - Three of a kind: 710
 - Full house: 95
 - Four of a kind: 44
 - Five of a kind: 4

Perform a Poker Test at $\alpha = 0.05$ level of significance to test independence (Critical Value $= 12.59$).

Paper 3: 2081 Chaitra

Attempt all questions.

1. [Q1] Define stochastic and deterministic models. Under what circumstances is simulation the appropriate tool, and when is it not appropriate?
2. [Q2] Explain the Acceptance-Rejection method for generating random variates with a suitable example.
3. [Q3] Differentiate between point estimation and interval estimation. How do you measure performance in a terminal simulation?
4. [Q4] Explain Markov chains and state transition probability matrices. Define absorbing states and steady-state probabilities.
5. [Q5] Perform a Kolmogorov-Smirnov (K-S) uniformity test with $\alpha = 0.05$ on the following $N = 5$ generated random numbers:

$0.44, 0.81, 0.14, 0.05, 0.93$

(Tabular Critical Value for $N=5$, $\alpha=0.05$ is $D_{\alpha} = 0.565$).

6. [Q6] Write short notes on:
 - (a) GPSS / SIMSCRIPT simulation programming languages
 - (b) Verification techniques for simulation software

Paper 4: 2079 Chaitra

Attempt all questions.

1. [Q1] What is system dynamic modeling? Discuss feedback loops with positive and negative causal loop diagrams.
2. [Q2] Explain multi-server queueing systems ($M/M/c$). Derive expressions for server utilization and probability of zero units in the system (P_0).

3. **[Q3]** Describe time-advance mechanisms: Fixed-time step vs. Next-event time advance with comparison diagrams.
4. **[Q4]** A barber takes exactly 30 minutes per haircut (deterministic service time D , service rate $\mu = 2/\text{hr}$). Customers arrive according to a Poisson process at a rate of 1 customer every 40 minutes (arrival rate $\lambda = 1.5/\text{hr}$). Find the average time a customer waits in the queue (W_q).
5. **[Q5]** Describe run length, warm-up period, and replication analysis in steady-state simulations.
6. **[Q6]** Write short notes on:
 - **(a)** Auto-correlation test for random numbers
 - **(b)** Monte Carlo simulation method